risk and uncertainty কিভাবে Handle করা হয়?

→ ছুটির পর স্যার Article থেকে Question করবেন ক্লাস-এ।

Date: 16-02-2022

Econ: 404

"সজীব স্যার"
কাওসার হামান সজীব [2021-এ February-তে Sir join করেছেন]

Asymmetric Information
Supply and Demand
Health Insurance Market

R.B: Folland

"সজীব স্যার" বই - 3টা chapter Sir পড়াবেন।

কাওসার স্যার - স্যার সব পড়িয়েছেন Folland এর বই থেকে। Slide ও দেখাবেন।

স্যার Slide, যা থেকে পড়াবেন সব Provide করবেন।

Asymmetric Information

Perfect Information: All consumers and producers have complete information on all prices services available in the market.

Imperfect information: Incomplete information to consumers or producers.

Asymetric information : Consumer or producer possess more information than other.

↓

level of information ও vary করে।

* Level of information will differ among participants much as between physician and patients.

* Asymetric information exist in all market more or loss

Q. How much asymmetric info. exists in Health sector?

Pauly (1978), one fourth or more of total personal health care expenditure might be regarded as "reasonbly

- Automobile - এর বেলায়
- Insurance market " "
- Health care "
- Electrician "

} Asymmetric Info. থাকে

informed". If we add nursing home services and chronic disease conditions, this ratio comes to one-third.

* Information gap exist within not only ~~par~~ patients but also physicians. Because uncertain outcome of an individi . . . . . .

— - - - . . - - - -

Mechanisms:

1) Certification
2) Accrediation
3) Threat of ~~m~~ malpractic
4) physician - patient - relationship
5) Ethical constraint

The Lemons Principle

Akerlof (1970)

Sellers of the used Car: He knows every information of cars available for auction.

0 থেকে 2 Quality
0 → low "
2 → highes "
1 → Average "

৩জন seller ; ৩টে Car.

Buyer of the used car: Avg. quality of each car.

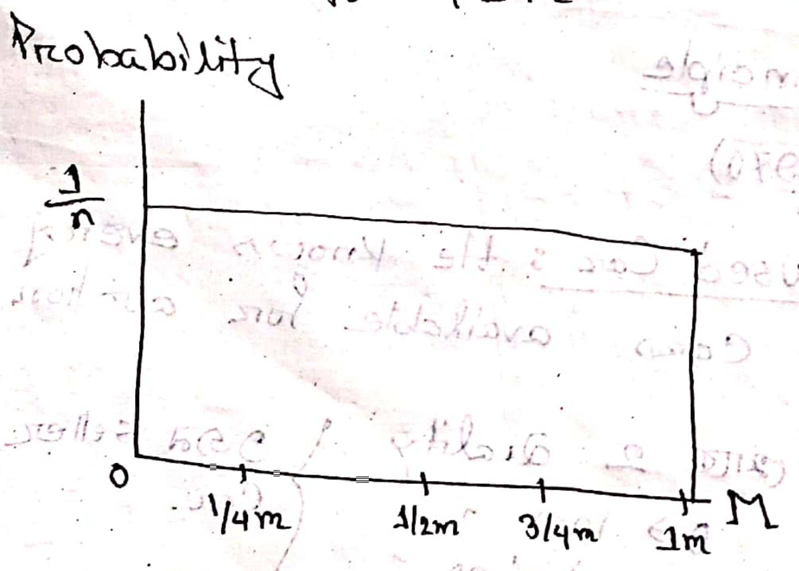
Avg. Q = 1Q   Low Q = 0Q

Hig Q = 2Q

Auctioneer: play the auction

- Application of Lemons principal in health insurance:

__Assume__: All patient have same demographic characteristics and their expected health expenditure level of the insured period range from \$0 to \$1m



Insurance company (buyer) = $\frac{1}{2}$M expenditure

Patient = he knows about expected expenditure

Auction = \$0 premium

$\downarrow \frac{1}{2}$M premium

[যখন Insurance company তাদের অফার এর $\frac{1}{2}$m এর above-এ আসে তা accept করে।

Imperfect Info.

level of Information between buyer and seller Same হবে But Complete হবে না।

Asymmetric Info.

level of Information same হবে না।
Buyer-এর কাছে এক রকম Info. থাকবে; Seller-এর কাছে এক রকম থাকবে।

① Inefficiency of adverse selection

~~Adverse S~~

② The affordable care act and adverse selection

③ Experience rating and Adverse selection.

Folland এর বইয়ে Ch-10-এ আছে। পরে আসতে হবে। Sir পড়াবেন না।

1st and 2nd Tutorial Syllabus

① Introduction to health economics,
Chapter 1

② Microeconomic tools for health economics
Chapter 2

③ Statistical tools for health economics
Chapter 3

④ Production of health
Chapter 5

⑤ Grossman Model.
Ch – 07

** Ch-1 ; Ch-05 and ch-07

Ch-02 ;

Pdf আছে

* Lec 06 – 10-12-20
* Lec 07 –
* Lec 08 –
* Lec 09 – 07-01-21
* Lec 11 –
* Lec 12 – 28-02-21
* Lec 13 – 02-03-21

Lec 15 –

Introduction of Health Economics: (Ch-1)

Q:1: What is health economics? Does Economics apply to health and health care?

Q:2: Why health care is different from other goods and services? Explain.

Q:3: Define health economics. Why does the share of medical care to GDP increase in recent years?

Q:4: Explain the relevance of health economics. Why do we study health economics?

Health economics is still a relatively new discipline with an evolving scope and pedagogy, and neither it, nor we, will provide answers to all the health system questions that nations face. Despite this caveat, we can not think of any field of study that is more relevant and has more urgently needed solutions.

Production of Health (Ch-05)

Q:5: Define health production function. Would the health production function eventually bend downward? Explain in support of your answer.

Q.6: Explain the two different theories of schooling. (বই থেকে পড়ছি)

Q.7: Describe the production function of health.

Q:8: The role of medicine in the historical decline of the population mortality rate is small, perhaps negligible. Explain this statement with empirical evidence

Q:9: According to Victor Fuch, what factors are responsible for the historical decline in the infant mortality rate? Explain. Does this mean medical research is unimportant. Explain.

Q:10: If not medicine, then what factors were responsible for the historical decline in the population mortality rate. Discuss.

Ch-02: Microeconomics Tools for Health Economics:

Q:11: Using supply and demand graph and assuming competitive markets, show and explain the effect on equilibrium price and quantity of the following:

(a) An increase of the price of Ace on the market for Napa (Hint. Ace and Napa are substitutes).

Q:12: (b) A price ceiling placed on physician fees on the market for physician services.

(c) Increased cost of medical education on the market for physician services.

(d) The virtual elimination of smoking in the population on the market for hospital services.

Ch-07: Demand for Health Capital

আসবে
Q: 12: * According to Grossman model, how health care demand is different from traditional approach of demand?

Q: 13: * Explain home good function and health investment function in the light of Grossman model.

Q: 14: How optimum health stock is determined? Explain with diagram

Q: 15: * Explain, in the light of Grossman model, how an individual chooses optimum amount of income and leisure.

Q: 16: * How is it possible to earn আসবে more money and more leisure time together? Explain, in the light of Grossman model.

Q: 17: → আসেনি Explain, in the light of Grossman model, the impact of age and education on the optimum health stock.

Q: 18: → আসেনি How would you explain the potential use of an individual's time by labor-leisure trade off? Show the equilibrium of an individual's time by labor-leisure line and indifference curve.

Q: 19: Explain the impact of an increase in activities to improve health on income and leisure.

Q: 20: What are the limitations of Grossman model?

Q: 11 এর বাকি অংশ:

(e) Increased graduations of new doctors on the market for physician services.

(f) Decreased price of pain killers on the market for anti ulcerative drugs.

(g) A technological change that reduces the cost of producing X-rays on the market for physician services.

It

Analyses

Q.11

(a) An increase of the price of ~~of~~ Ace on the market for Napa.

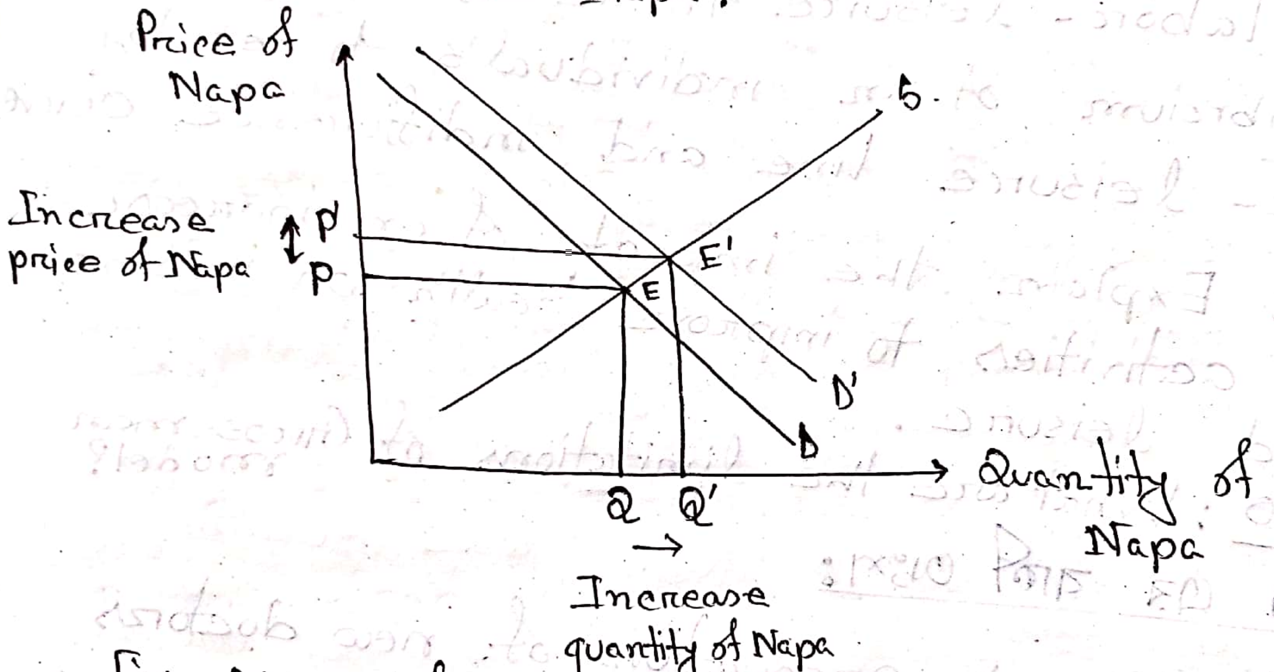


Fig: Market for Napa

Since Ace and Napa are substitute goods. An increase of the price of Ace will increase the demand for Napa. The Demand Curve shift from D to D'. The quantity demanded for Napa increases from Q to Q', and price increases from P to P'.

The equilibrium position of the competitive market shifted from E to E'.

(b) The price ceiling placed on physician fees on the market for physician services

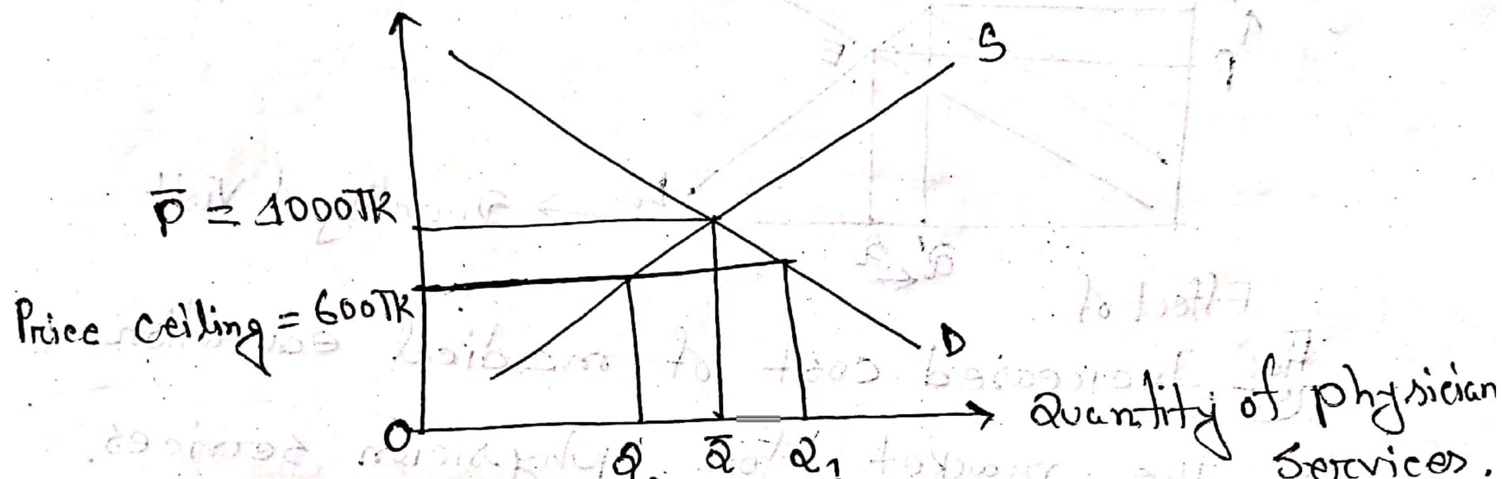


Fig: Market of physician services.

Without price ceiling the quantity of visit was $\bar{Q}$. Let $\bar{P}$ = 1000 tk price of per visit. Suppose govt. placed a price ceiling on physician fees of 600TK. This creates an excess demand for physician service, Q_1. But the supply of the physician services decreases to Q_0. As the demand for physician services increases and supply decreases, this decrease the quality of physician service.

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(c) Increased cost ~~for~~ of medical education on the market for physician services.

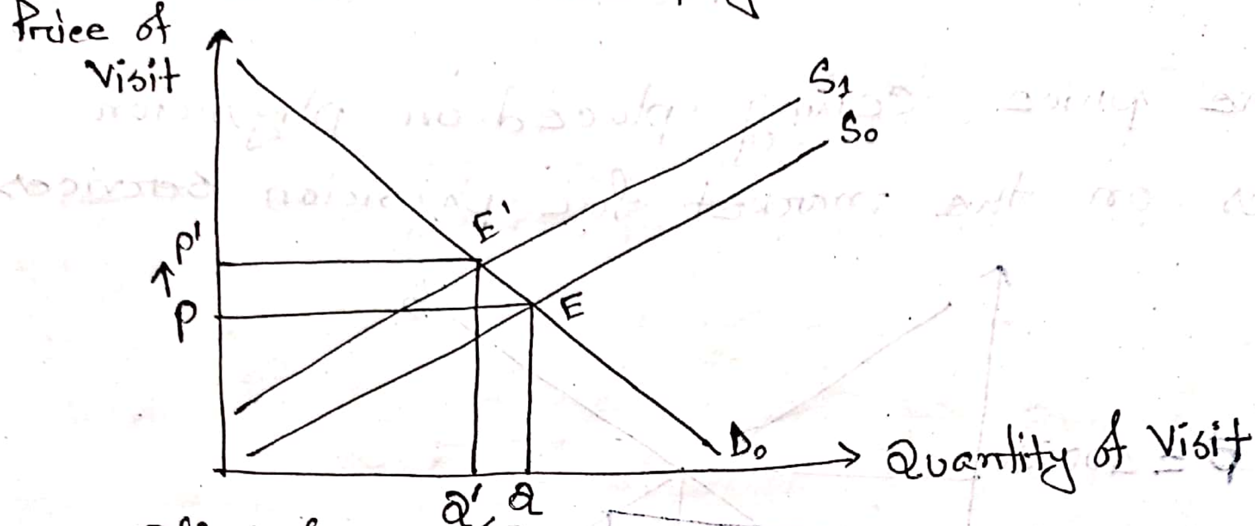
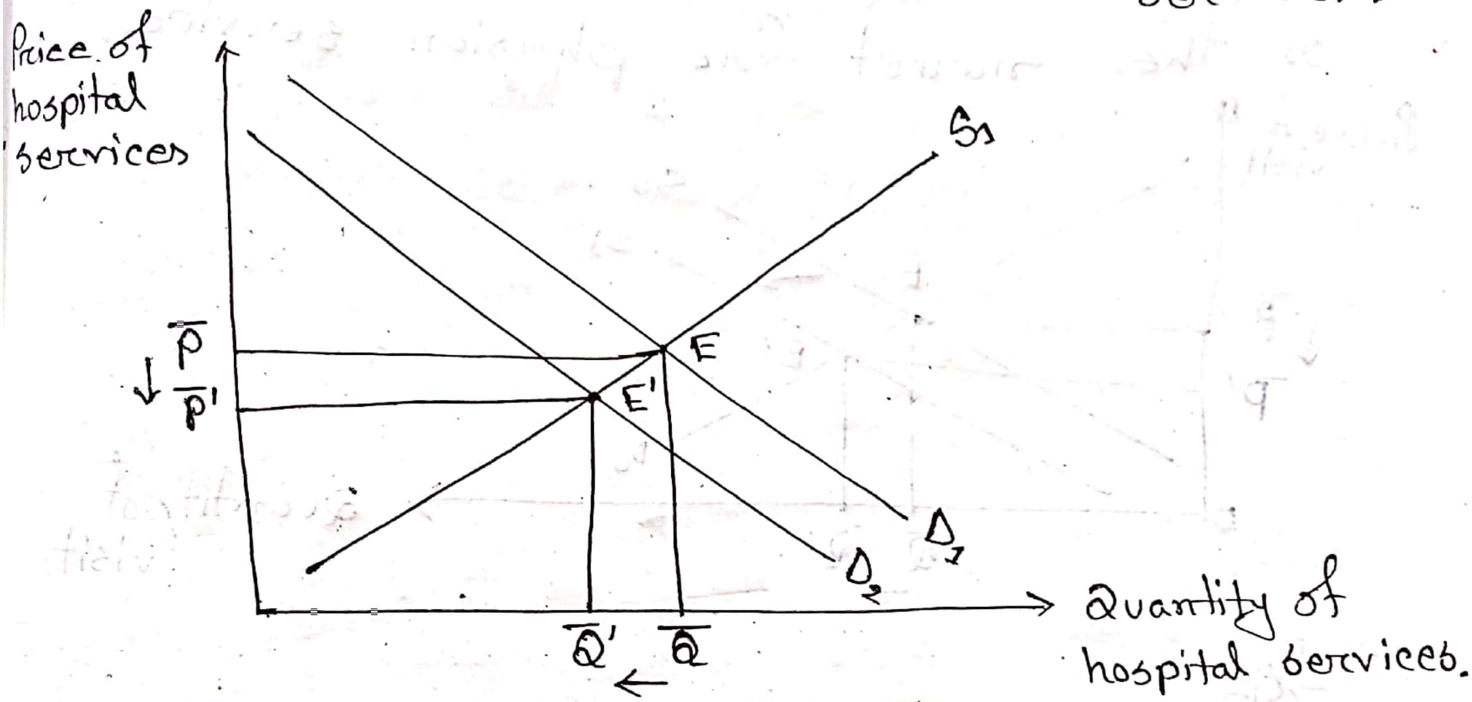


Fig: Effect of Increased cost of medical education on the market for physician services.

If the cost of medical education increases, It will lead to decrease in the supply of of physician services. Because of high cost of medical education, high price of medical services is charged.

In the above figure, the supply curve shifts from S_0 to S_1. So, the price of visit increases from P to P' and the quantity of visit decreases from Q to Q'. The equilibrium position shifted from E to E'.

(d) The Virtual elimination of smoking in the population on the market for hospital services.



As a result, the
(When people smoke less then) demand for hospital services decreases from D_1 ~~$\bar{Q}$~~ to D_2 ~~$\bar{Q}'$~~. therefore price for hospital services decreases from $\bar{P}$ to $\bar{P}'$ and also quantity of hospital services decreases from $\bar{Q}$ to $\bar{Q}'$.
The equilibrium position on the market for hospital services shifts from E to E'.

→ The virtual elimination of smoking will improve people's health status so ~~that~~ people now going less to doctors.

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(e) Increased graduations of new doctors on the market for physian service.

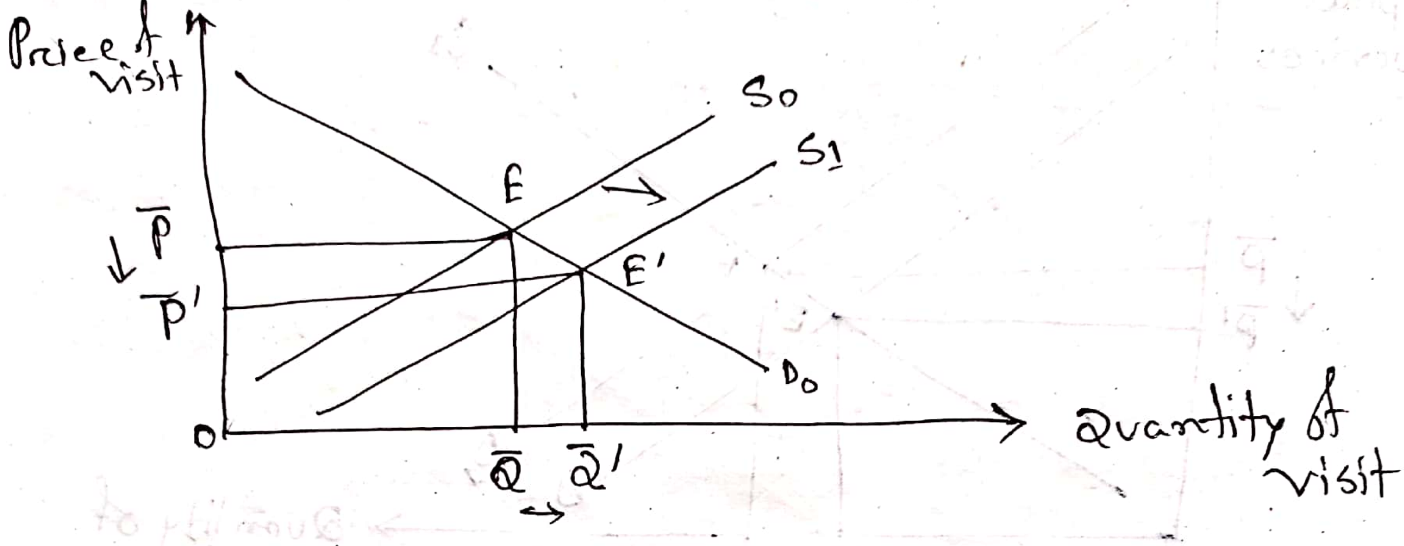


Fig:

Increased graduations of new doctors on the market for physician services will shift the supply curve to the right from S_0 to S_1. As a result equilibrium also shifts from E to E'. Now equilibrium price of visit decreases from $\bar{p}$ to p' and quantity of visit increases from $\bar{Q}$ to $\bar{Q}'$.

(8) ~~Decreased~~ Increased price of pain killers on the market for anti ulcerative drugs.

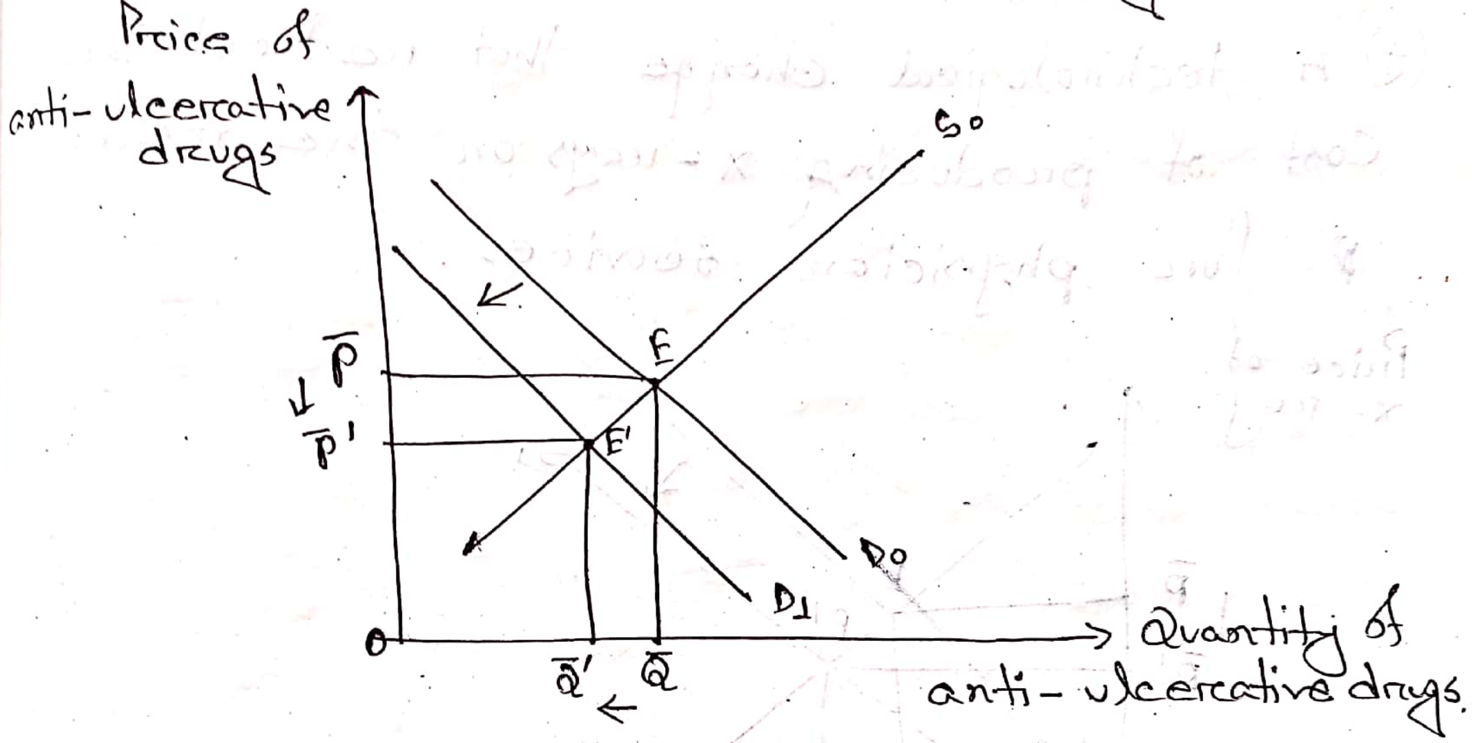
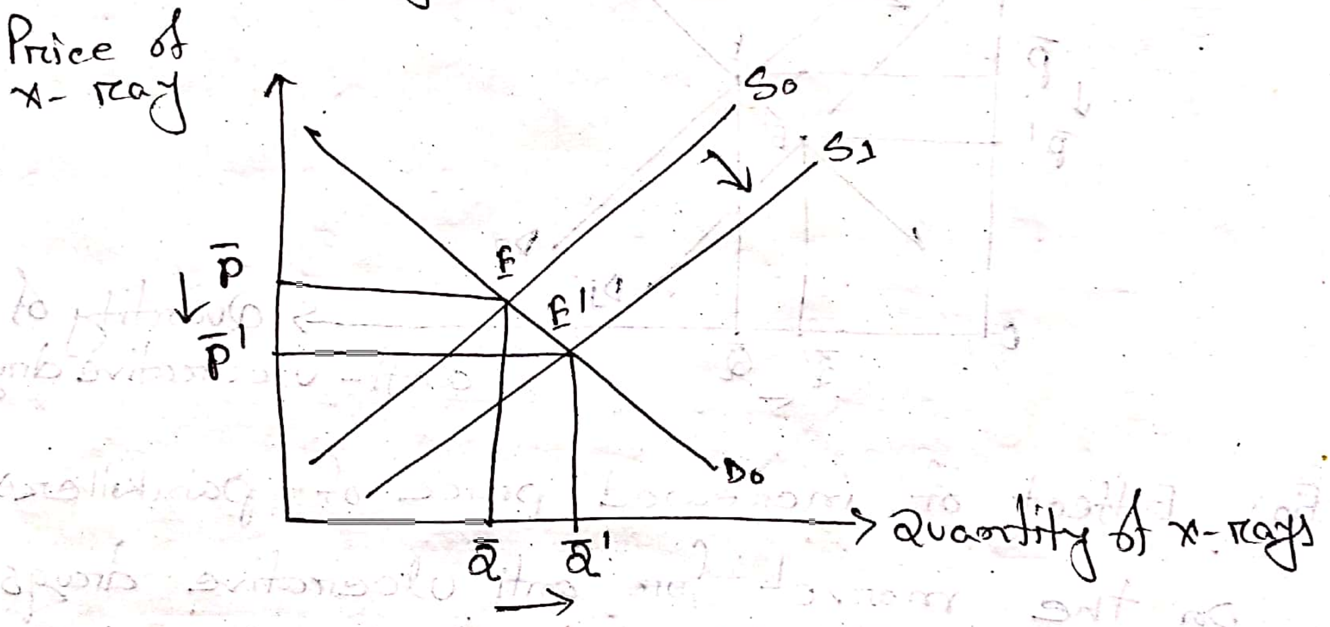


Fig: Effect of increased price of painkillers on the market for anti ulcerative drugs.

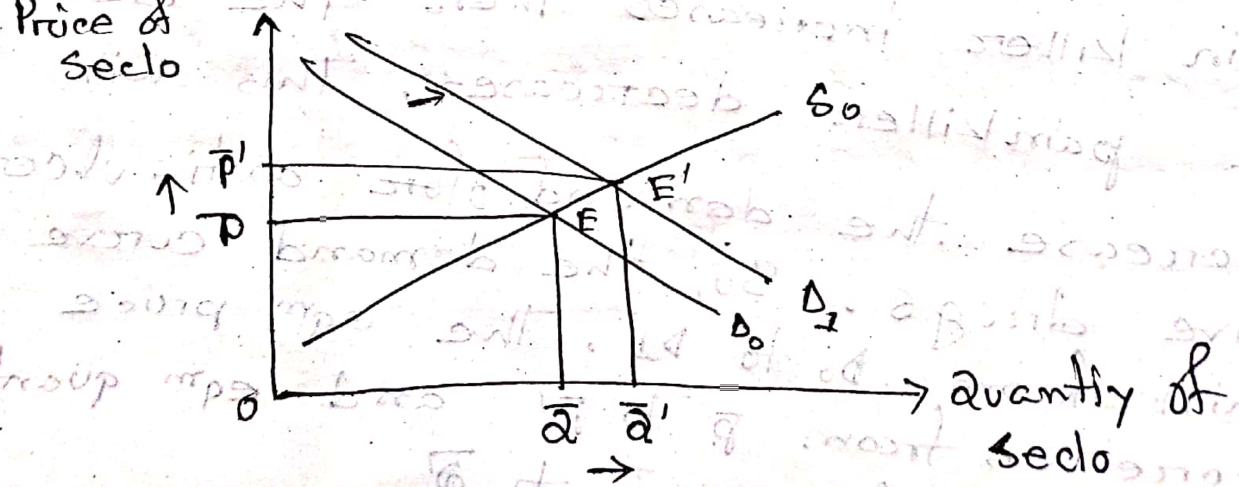
Pinkiller and anti ulcerative drugs are Complementary goods. When prices of pain killer increases then the demand for painkiller decreases. This will decrease the demand ~~of~~ for anti-ulcerative drugs. So, the demand curve shift from D_0 to D_1. The eqm price decreases from $\bar{P}$ to $\bar{P}'$ and eqm quantity decreases from $\bar{Q}$ to $\bar{Q}'$.

The new eqm shift from E to E'.

(g) A technological change that reduces the Cost of producing x-rays on the market for physician services.



(h) Decreased price of Rolac on the market for Seclo (Rolac and Seclo are complements)



Answer to the question of (05)

Production function of health:

Production function of health refers to the functional relationship between health care inputs and health status.

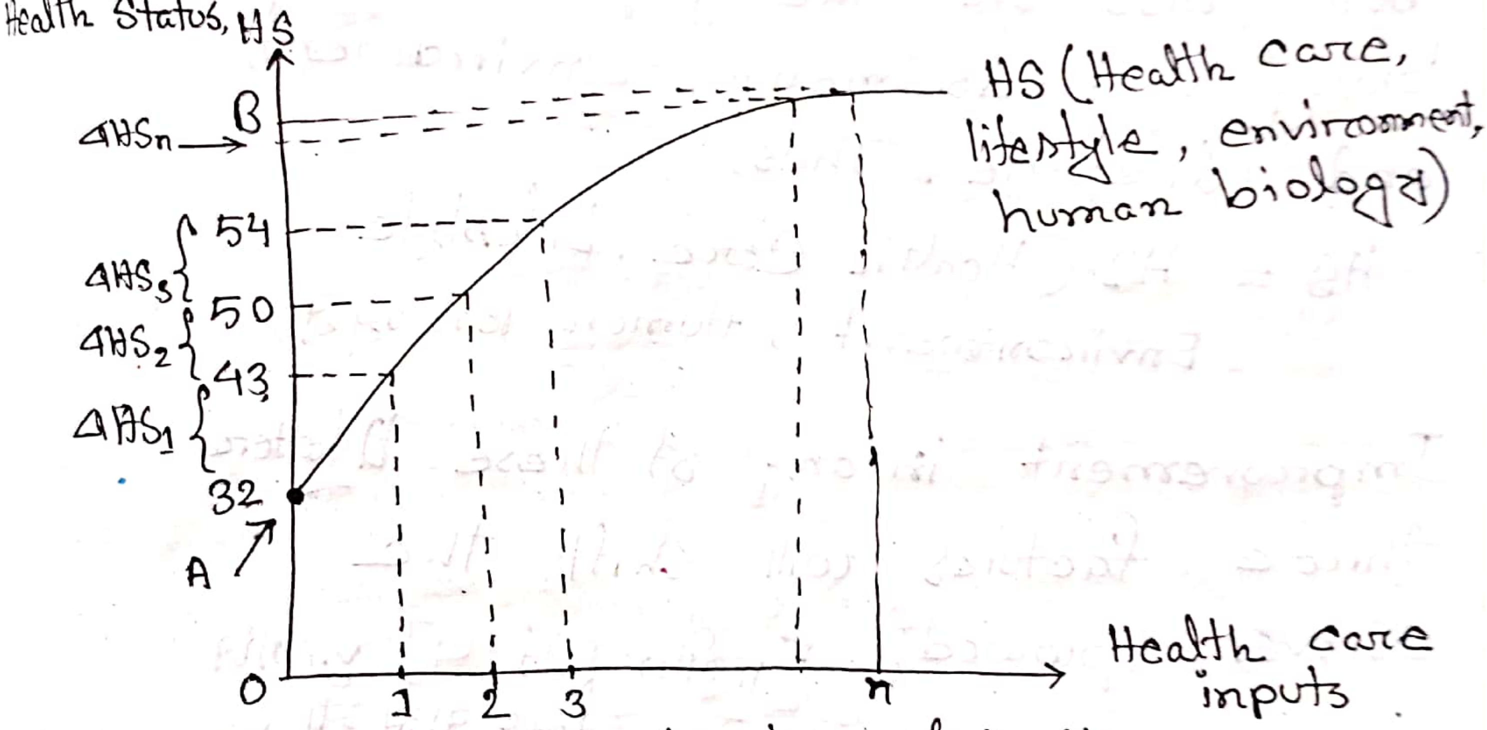


Fig 1: Production Function of Health

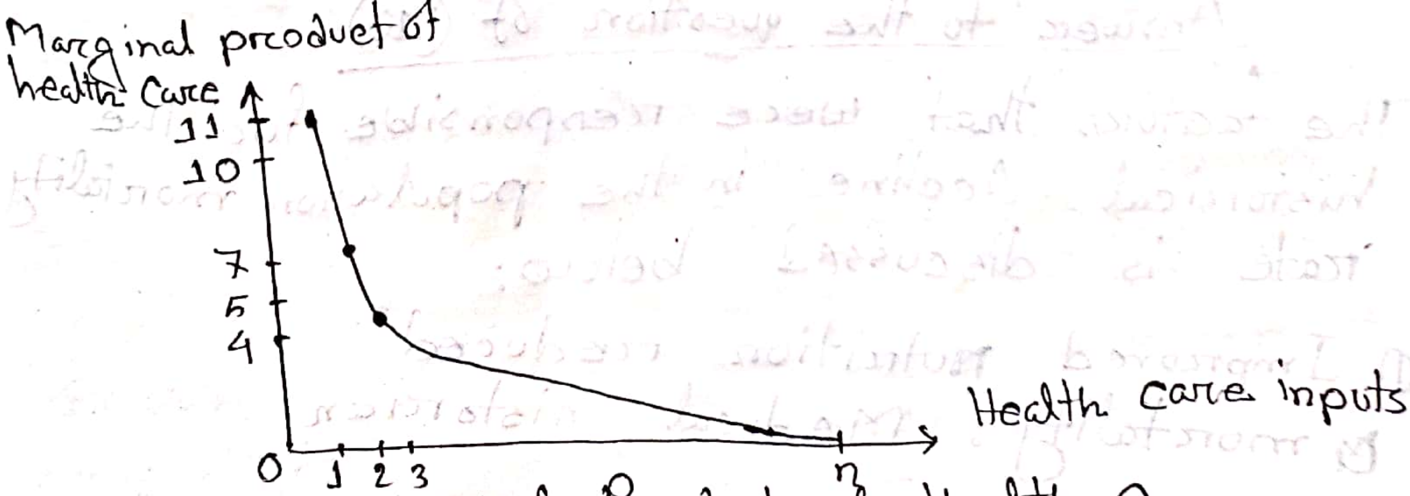


Fig 2: Marginal Product of Health Care.

In fig (1) health status is shown to be an increasing function of health care. Actually in general a person's health status depend's on not only on health care but also on the population's biological endowment, environment, and lifestyle. Thus,

HS = HS (Health Care, Lifestyle, Environment, Human Biology)

Improvement in any of these latter three factors will shift the curve upward. [বাকীটা pdf-৭] হুবহু লিখলাম না।

Answer to the question of (10)

The factors that were responsible for the historical decline in the population mortality rate is discussed below:

① Improved Nutrition reduced mortality: Medical historian

Thomas McKeown and economic historian Robert Fogel argued strongly that the main cause was improved nutrition. Robert Fogel provided the needed evidence. He established that after the mid-eighteenth century, calorie intake of Europeans increased tremendously.

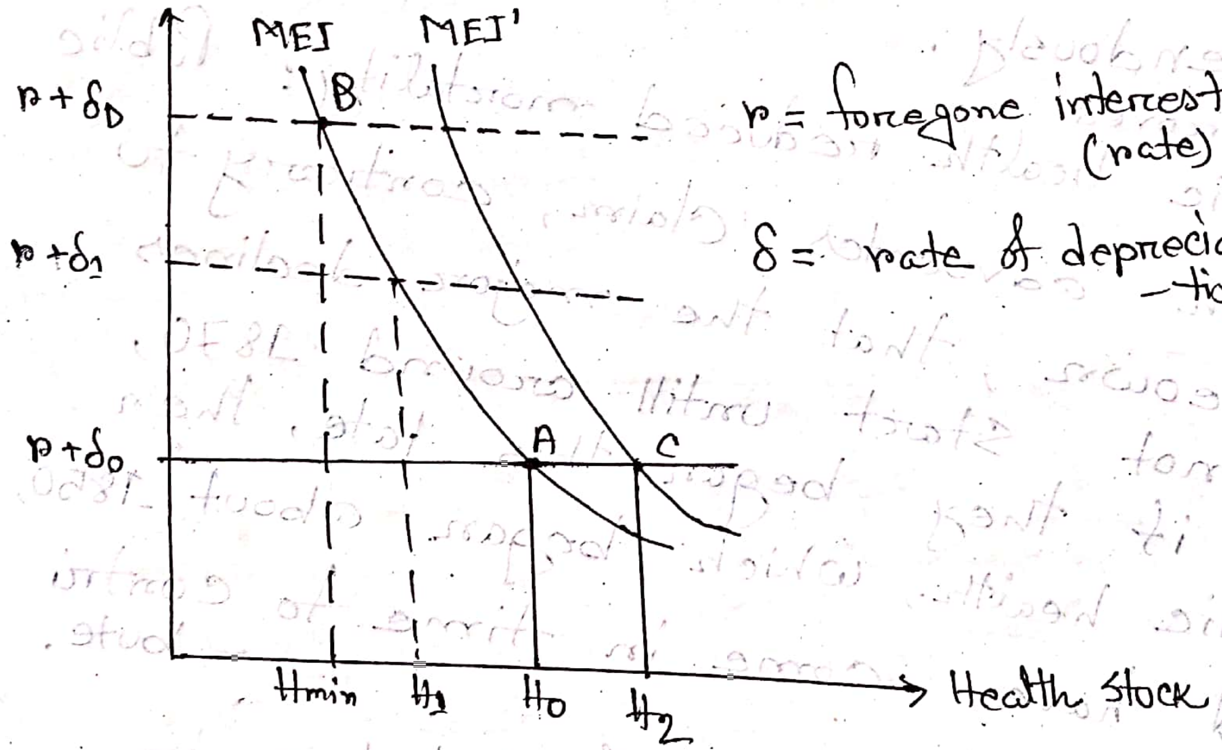
Improved Public Health reduced mortality: Public

(17) Improved Public Health reduced mortality: Public health advocates claim, contrary to McKeown, that the major declines did not start untill around 1870, and if they began this late, then public health, which began about 1850, would have come in time to contribute.

~~Moreover~~, A number of scholars in the field of Health economics attribute the largest share of the credit to ~~improve~~ decline population mortality rate to improvements in environment, particularly to the greatly increased supply of foodstuffs that became available due to the agricultural and industrial revolutions.

Optimal Health Stock

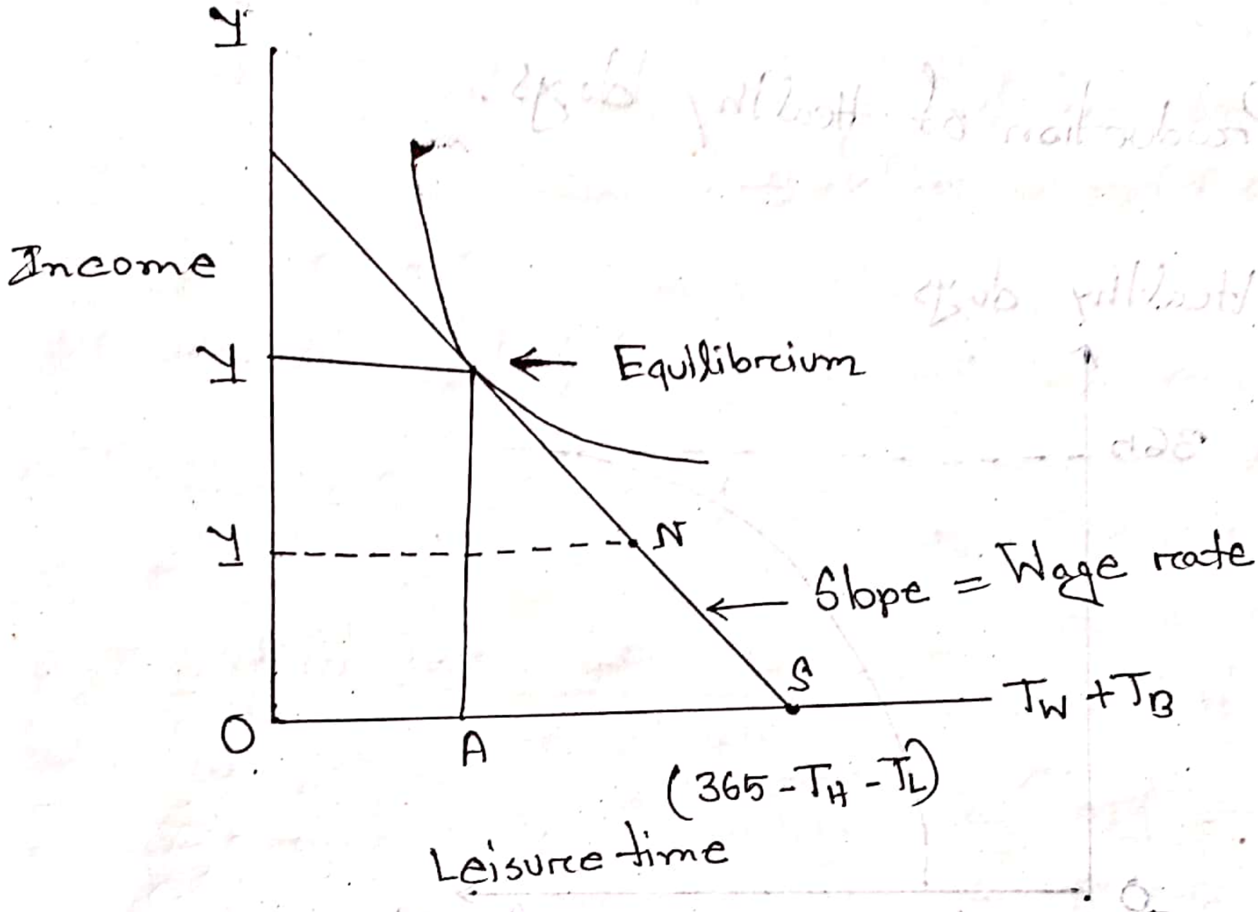
Cost of Capital



r = foregone interest (rate)

δ = rate of depreciation

Labor - Leisure trade - off



Preferences between Leisure and Income

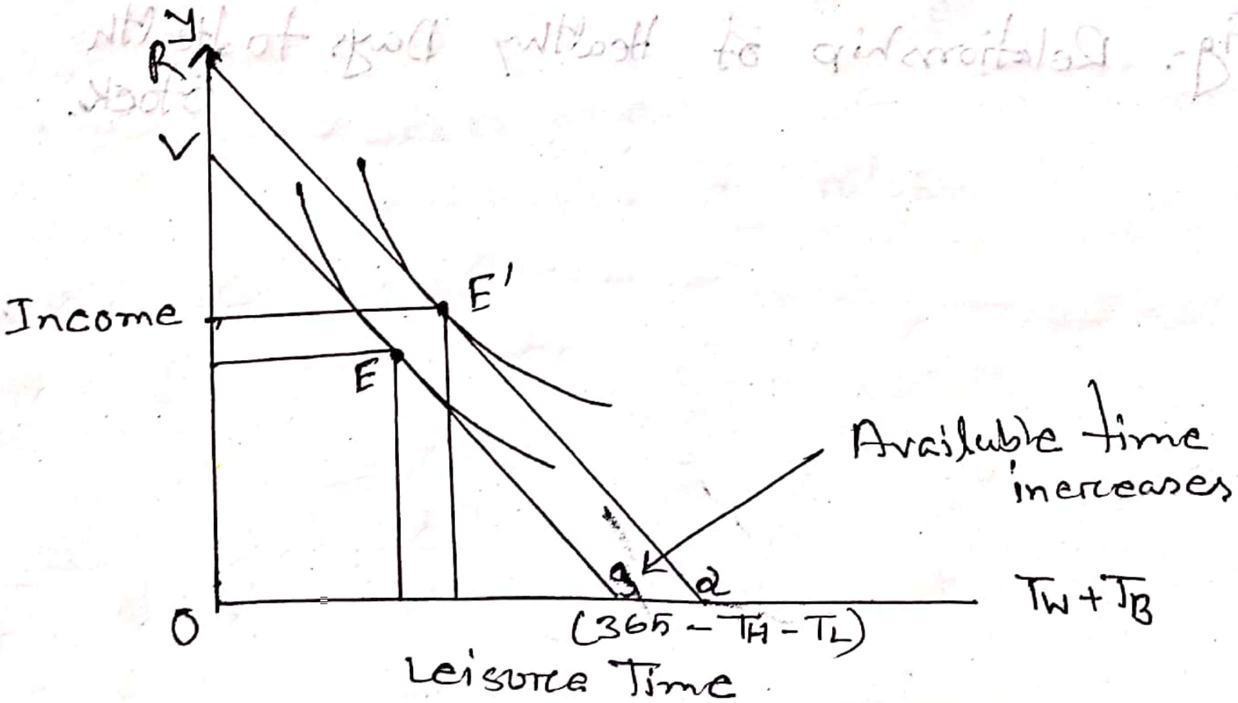


Fig: Increased Amount of Healthy Time Due to Investment.

Production of Healthy days.

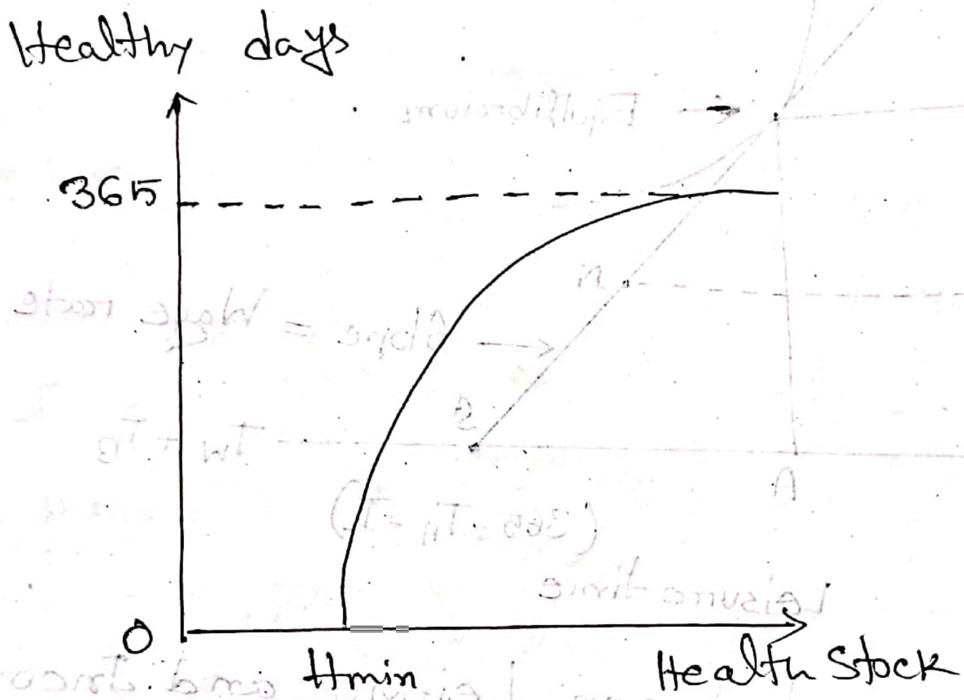


Fig: Relationship of Healthy Days to Health Stock.

Econ: 404 (Health Economics)   Date: 23-02-22

"অতীব স্যার"

Inefficiencies of Adverse Selection:

Under insure → low risk কিন্তু বেশি পরিমাণ Premium দিতে হয়।

Over insure → high risk " কম " " "

এই 2টা Inefficiency তৈরী করে।

If the lower risks are grouped with higher risks and all pay the same premium, the lower risk face an unfevourable rate and will tend to under insure. Conversely, the higher risk will face a favourable premium and therefore over insure.

Thus, in addition to inefficiency, income will be re-distributed from consumers who are lower risks to those who are higher risks

Reason behind this gap:

1. Asymmetric information
2. lowering the price of health care to the patient ~~ever~~ encourages the

Consumption of additional health care.

③ The insured may have different observable characteristics such as age, and other.

50% more expenditure [Insured vs Uninsured]

↓

Insured - এর হয় তেল উপরে ৬টা gap (সমস্যাগুলো) হল।

ACA and Adverse Selection

[ACA = Affordable Care Act]

2014, ACA mandatory

→ Tax penalties
→ Federal subsidies
→ Health insurance exchage

→ ACA না নিলে এই ৩টা জিনিস দিতে হবে।

18 - 34 → young → \$1700/y
→ \$350/y [খরচ হয়]

\$300 + \$500 = \$850 < \$1700

↙

tax ~~50~~ + Fed sub.

Experience Rating and Adverse selection:

Group insurance → [যখন কোন Organization or employer তার employee দের জন্য Insurance খুলে]

Group insurance to implement experience rating, where premiums are based on the past experience of the group or the other risk rating systems to project expenditures.

Health Maintenance Organization (HMO)

Agency Relationship

Principal : Patient

Agent : Service provider (Doctors, HC Facility, DL)

Perfect Agent : Physician who chooses as the patients themselves would chose if only the patients possessed the information that the physician does.

: "Medical Code of ethics".

Dranove and White (1987)

Why we do not reimburse physician based on the basis of improvement?

Asymmetric Information : Patient's improvement information also possessed by the patient him/herself. He or she may or may not disclose this information.

4 Consumer information and prices

1981 এর একটা study. করেছেন Pauly and Satterthwaite

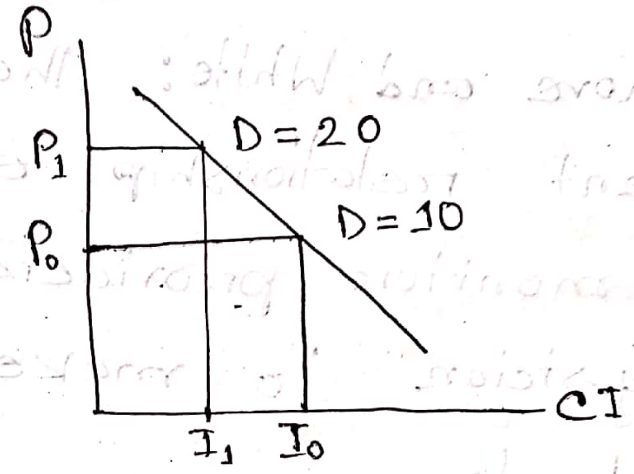
Reputation good : primary medical care

→ Friends
→ Neighbours
→ others

50
↓
5 person → 1 Doctor

(10) জন Doctor

(20) 〃 〃



Role of informed buyer : Price determiner for the rest of uninformed buyer.

Consumer information and Quality :

~~McGly~~ McGlynn and Colleages (2003), evaluated the medical records over a two year period for a random sample of adults in 12 metropolitan areas of USA.

Findings :

Only 55% received recommended.

Services :

1. Preventive Care 55%
2. Accute Care 54%

③ Chronic Condition Care 56%

Dranove and White: The physician - patient relationship enables patients to monitor providers and encourages physician to make appropriate refferals.

[Assymmetric Chapter - (মধ্যে)]

Econ: 404 সজীব স্যার Date: 27-02-22

Supply Induced Demand (SID)

সব সময় SID খারাপ না।

Q: What is SID? Is it always bad for principal?

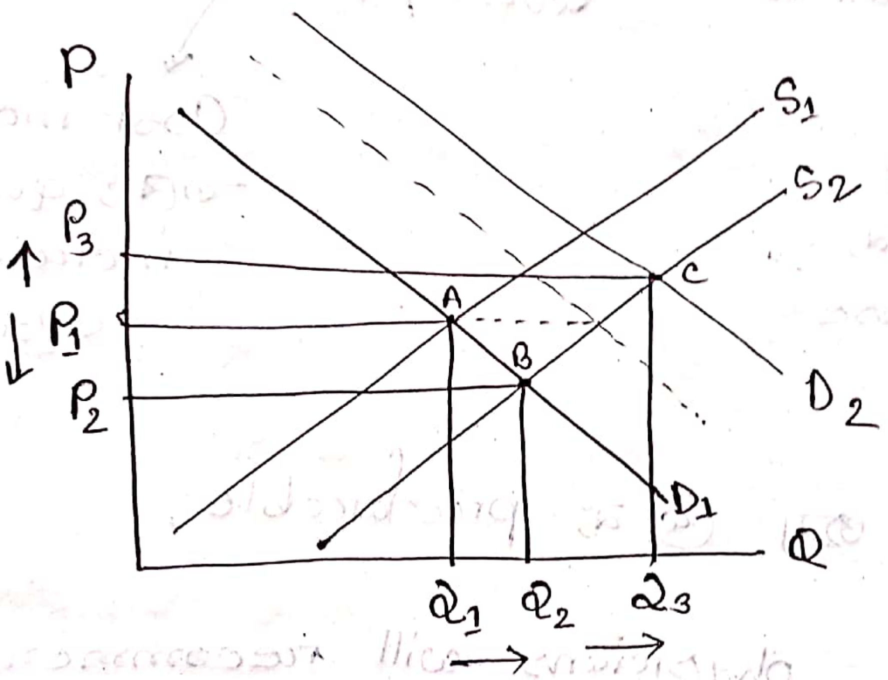
→ Asymmetric Information

- agent → physician
- Principal → Patent

Violation of agency

কিন্তু Incentive physician Patient-কে provide করে যা Patient এর জন্য খারাপ না।

Supply and Demand Model: ~~When this SID~~



→ এই situation (SID) ভালো।

* When this SID isn't good:

① Motive or incentive of physician is to make more profit

ⅱ ~~Princi~~ Principal or patient gets wrong treatment.

Q. Do physician respond to profit incentives:

a. pay per service → Quality ↑

b. Capitation → Quality ↓

শুধুমাত্র Cost increase করে, Quality increase করে না।

Cost increase করে; qualitio increase করে।

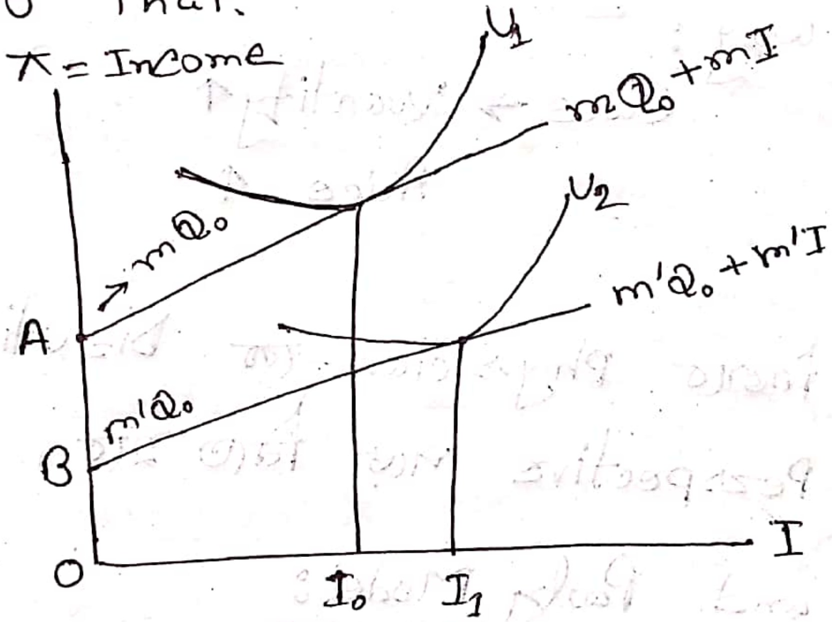
Patient - এর জন্য ক নং preferable.

① OB/GYN physicians will recommend C. section when their practice incomes are threatend by competition

② Patients whose physicians receives reimburnsement under a capitation systems may get fewer services and low in quality.

③ Physicians have personal goal for income.

④ When Govt. intervene and ~~pri~~ provide incentives for physician to ~~Close~~ chose more cost Saving methods, physicians respond to that.



Assume:

Profit rate = m, m'

Service Provided = Q

Induced Quality = I

Income = π

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Date: 02-03-22

Supply Induced Demand (SID)

Target Income Hypothesis:

physician → Target Income → 500/=

Actual Income → 400/=

100/=

100/= বাড়ানো way:

Care → Quantity ↑

Price ↑

- Inducement করাতে Physician-কে Disutility দেয়?
- Physician Perspective থেকে নিতে হবে।
- McGuire and Pauly Model:

Utility maximizing physician decision:

1. Net income (Y) ⎫
2. Leisure (L) ⎭ → Positive Utility
3. Inducement (I) → Disutility

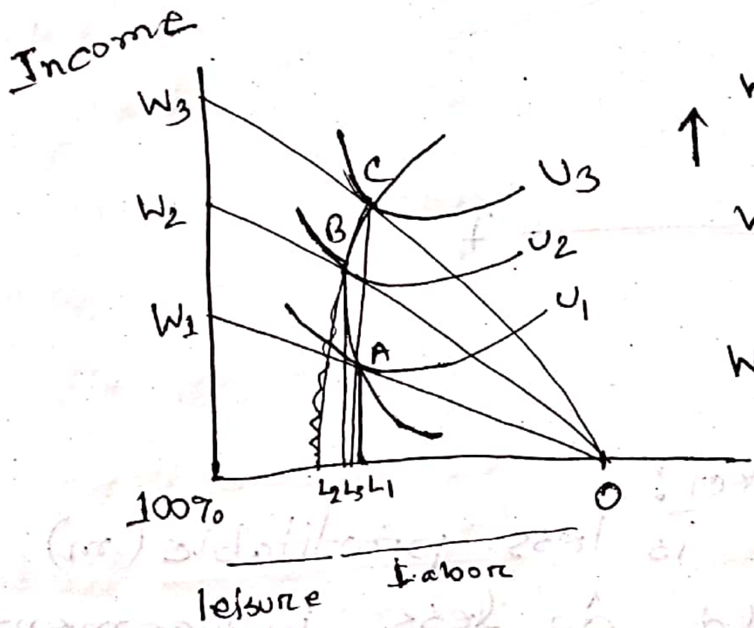
$U = U(\pi, L, I)$

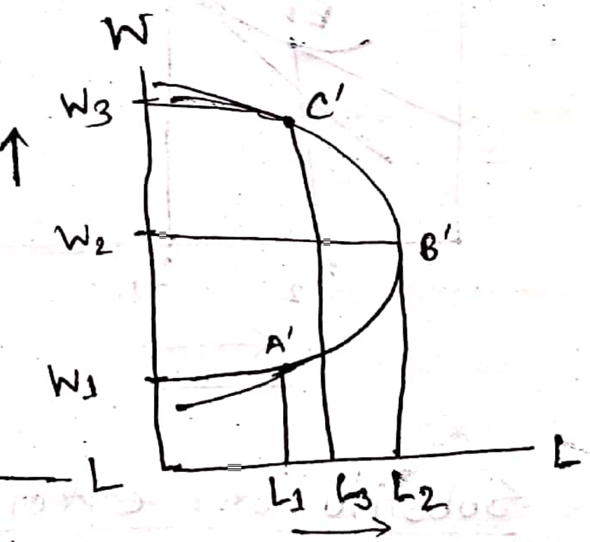
Pair Trade-off

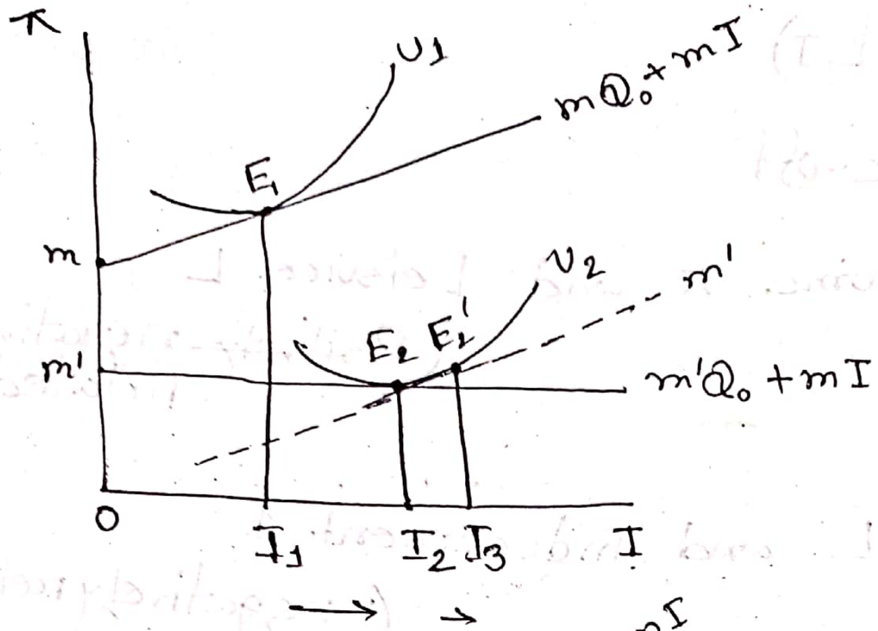
① Net income π and Leisure L
(~~Positively~~ negatively related)

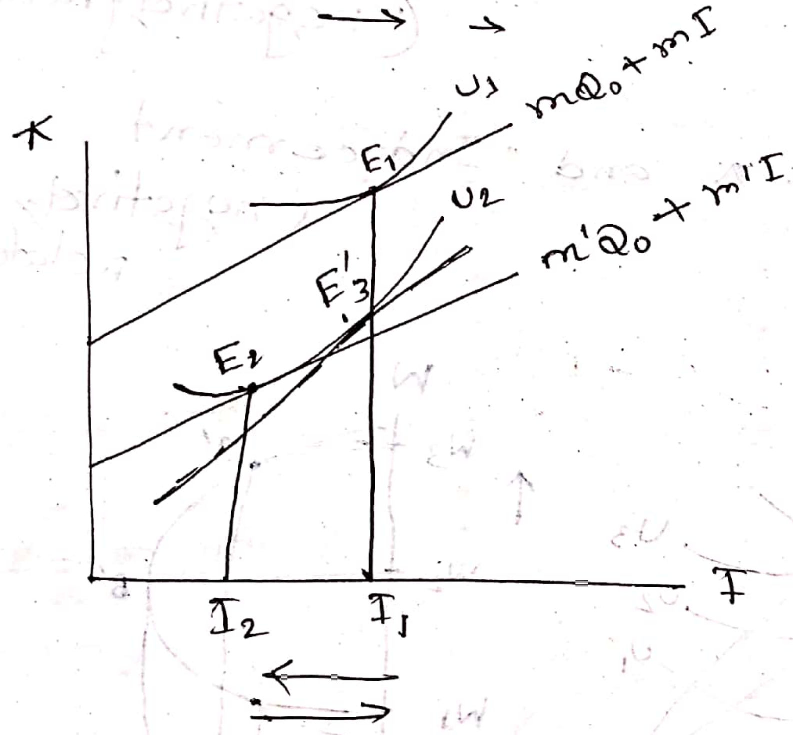
② Leisure L and Inducement I
(negatively related)

③ Net Income π and Inducement
(negatively related)









Substitution effect:

If inducement is less profitable (m) provider would do less inducement and substitute it away.

Income effect: Decreased income would make inducement more ~~desiarable~~. desireable.

Econ: 404

Ref: Ch 11 Folland

Date: 06-03-22

The Organization of Health Insuanance Market

Loading Cost:

Characteristics of Insurance Market in perfect competition:

$\pi = 0$

① $\pi > 0 \rightarrow$ Enter and $\pi < 0$ Exit

② $\pi = 0 \rightarrow$ ~~Case~~ Cease of Entry and Exit

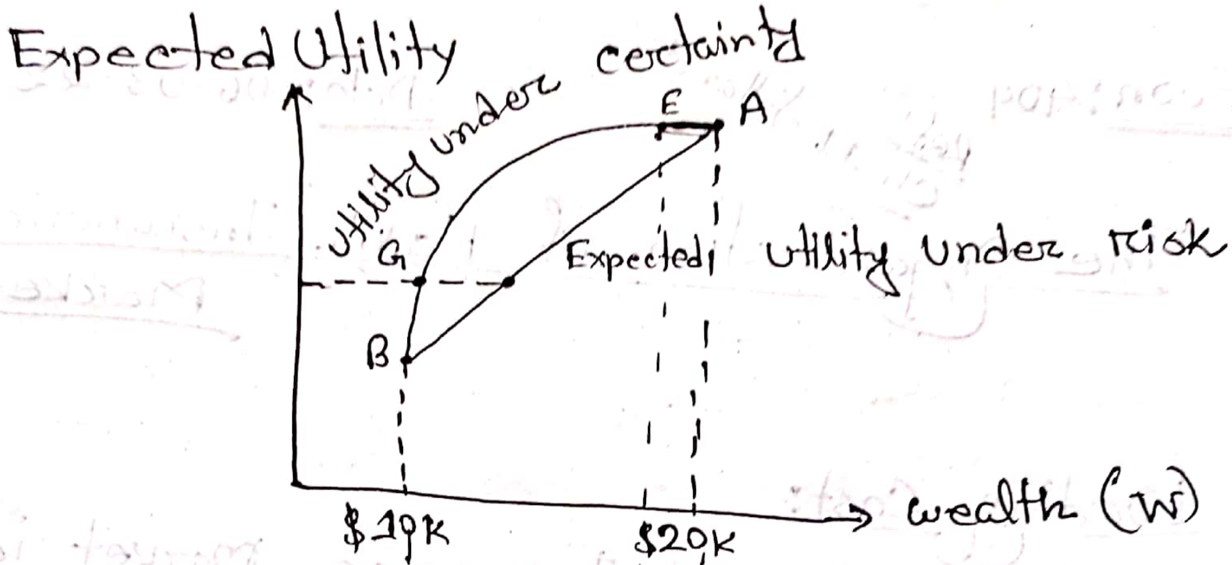
③ In good years $\rightarrow$ Claim pay out < premium collected $\rightarrow \pi > 0 \rightarrow$ Entry

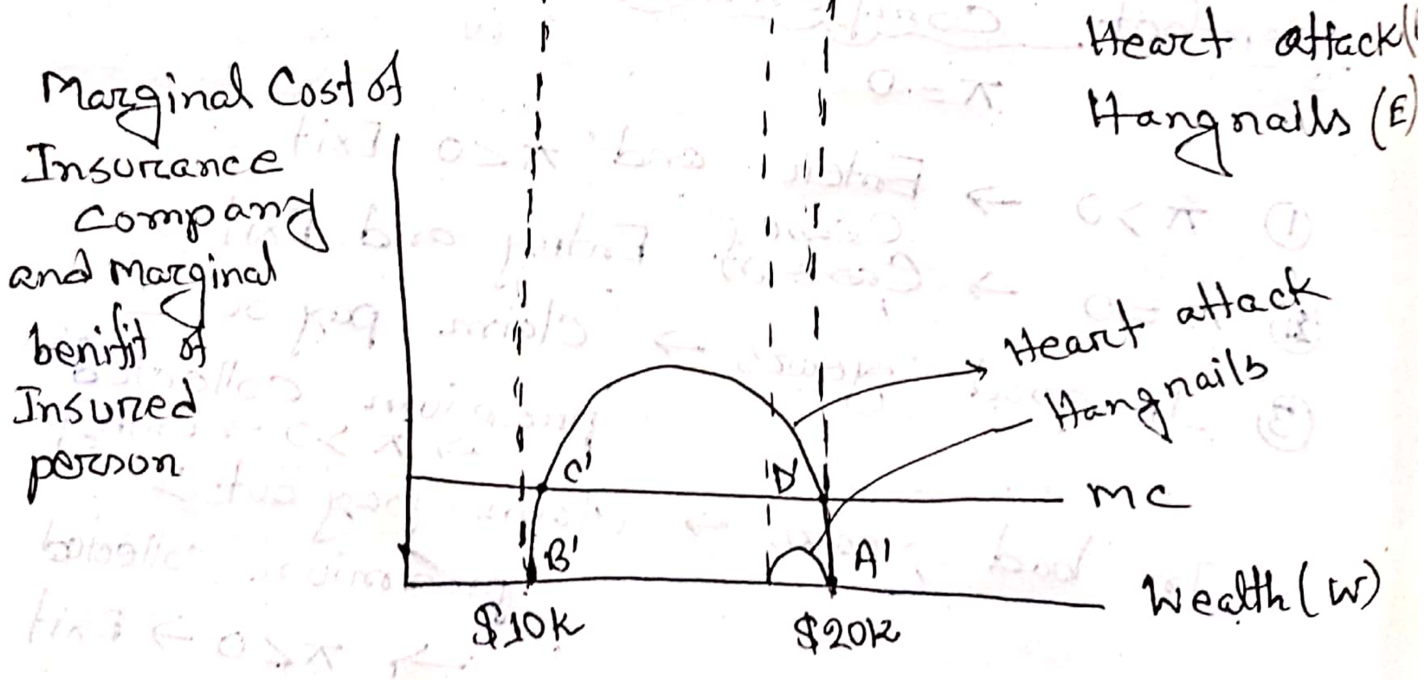
In bad years $\rightarrow$ claim pay out > premium collected $\rightarrow \pi < 0 \rightarrow$ Exit

④ Insurance company reluctant to cover price - elastic health care services.

Impact of Loading Cost

Utility of wealth under certain and risk associated





$A \to E \to$ Small loss

$A \to B \to$ Greater loss

Probability 1 হলে certain → Insurance নিবে না।

" 0 " → " " "

Group insurance:

→ Demography

→ Health reciept

→ History

→ Operational Cost

Employer provision of Health Insurance:

Who pays?

World War II